CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. EE Sem – I Examination, 2012-13
EE - 701 Engineering Optimization Techniques

Date: 18/12/2012, Tuesday **Time: 10 to 1 p.m** **Maximum Marks: 70**

Instructions:
1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

Q-1 **A.** Use graphical method to solve following Linear Programming Problem
Max. $Z=x_1+2x_2$
s.t. $x_1-x_2\leq 1$, $x_1+x_2\geq 3$, $x_1,x_2\geq 0$ — 7

B. Solve the following LPP using two-phase method **OR** simplex method
Max. $Z=3x_1-x_2$
Subject to: $2x_1+x_2\geq 2$, $x_1+3x_2\leq 3$, $x_2\leq 4$, $x_1, x_2\geq 0$ — 7

Q-2 **A.** Define: Dual variables, and obtain dual of the following primal problem — 7
Min $Z=2x_1+3x_2+4x_3$
s.t. $2x_1+2x_2+3x_3\leq 4$, $3x_1+4x_2+5x_3\geq 5$, $x_1+x_2+x_3=7$, $x_1,x_2,x_3\geq 0$

B. Using mathematical equations and graphs, explain difference between Exterior penalty function method and Barrier method — 7

Q-3 Define: basic variable. Solve the following sum using Big-M method — 7
Max. $Z=6x_1-3x_2$
Subject to: $x_1+x_2\leq 1$, $2x_1-x_2\leq 1$, $-x_1+2x_2\leq 1$, $x_1,x_2\geq 0$

OR

Q-3 Explain the term "Robustness" for the optimization algorithm. Write algorithmic steps of an Interior point method for linear programming. — 7

SECTION – II

Q-4 **A.** Find out minimum value of the given function using Kuhn-Tucker conditions — 7
Min $f(x_1, x_2)= (x_1-1)^2+(x_2-5)^2$
Subject to: $-x_1^2+x_2\leq 4$, $-(x_1-2)^2+x_2\leq 3$

B. Define and explain convex function and convex set — 7

Q-5 **A.** Using sequential quadratic programming **OR** Non-linear programming, minimize following function: — 10
Perform only two iterations.
Min $Z=-2x_1-x_2+x_1^2$
Subject to: $-2x_1-3x_2\geq -6$, $-2x_1-x_2\geq -4$, $x_1, x_2\geq 0$

B. Define: Linearly dependent vectors and shadow price — 4

Q-6 **A.** Write algorithmic steps for Quasi-Newton method **OR** Newton's method. — 5

B. Define: stationary points — 2
